

ASSIGNMENT #12 & #13 December 6, 2022

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SECTION 10.1

Exercise 1 (10.1.11). Let M be the abelian group $\mathbb{Z}/24\mathbb{Z} \times \mathbb{Z}/15\mathbb{Z} \times \mathbb{Z}/50\mathbb{Z}$.

(a) Find the annihilator of M in $\mathbb{Z}$ (i.e, a generator for this principal ideal).

Proof. Consider the ideal generated by the $l = \text{lcm}(24, 15, 50)$. We will proceed by showing that $(l) = \text{ann}_{\mathbb{Z}}(M)$. Suppose $a \in (l)$ so $a = lj$ some j multiple of l in $\mathbb{Z}$ and let $m \in M$ such that $m = (a, b, c)$ where $a \in \mathbb{Z}/24\mathbb{Z}$, $b \in \mathbb{Z}/15\mathbb{Z}$ and $c \in \mathbb{Z}/50\mathbb{Z}$. Then note that since $l = \text{lcm}(24, 15, 50)$ we get $am = (lja, ljb, ljc) = (0, 0, 0)$. So $a \in \text{ann}_{\mathbb{Z}}(M)$ and $(l) \subseteq \text{ann}_{\mathbb{Z}}(M)$.

Let $j \in \text{ann}_{\mathbb{Z}}(M)$. Then by definition for all $m = (a, b, c) \in M$ we get that $jm = (ja, jb, jc) = 0$. So it must follow that that $j(1, 1, 1) = (j, j, j) = (0, 0, 0)$ so it must be the case that $j \in (\text{lcm}(24, 15, 50)) = (l)$. Therefore $\text{ann}_{\mathbb{Z}}(M) \subseteq (l)$, thus finally $(l) = \text{ann}_{\mathbb{Z}}(M)$ □

(b) Let $I = 2\mathbb{Z}$. Describe the annihilator of I in M as a direct product of cyclic groups.

Proof. □

SECTION 10.2

Exercise 2 (10.2.4). Let A be any $\mathbb{Z}$ -module, let a be any element of A and let n be a positive integer. Prove that the map $\varphi_a : \mathbb{Z}/n\mathbb{Z} \rightarrow A$ given by $\varphi_a(\bar{k}) = ka$ is a well defined $\mathbb{Z}$ -module homomorphism if and only if $na = 0$. Prove that $\text{Hom}_{\mathbb{Z}}(\mathbb{Z}/n\mathbb{Z}, A) \cong A_n$, where $A_n = \{a \in A \mid na = 0\}$.

Proof. Let A be any $\mathbb{Z}$ -module, let a be any element of A and let n be a positive integer. Suppose the map $\varphi_a : \mathbb{Z}/n\mathbb{Z} \rightarrow A$ given by $\varphi_a(\bar{k}) = ka$ is a well defined $\mathbb{Z}$ -module homomorphism. Note that $na = \varphi_a(\bar{n}) = \varphi_a(\bar{0}) = 0a = a$. Now suppose $na = 0$ and let $i \cong k \pmod n$, so by definition we know that there exists some $j \in \mathbb{N}$ such that $i = nj + k$. Note that $\varphi_a(\bar{i}) = ia = (nj + k)a = nja + ka = j(na) + ka = ka = \varphi_a(\bar{k})$. Therefore φ_a is well-defined.

Consider the mapping $\pi : A_n \rightarrow \text{Hom}_{\mathbb{Z}}(\mathbb{Z}/n\mathbb{Z}, A)$ defined by $\pi(a) = \varphi_a$. Suppose $\varphi_i, \varphi_j \in \text{Hom}_{\mathbb{Z}}(\mathbb{Z}/n\mathbb{Z}, A)$ such that $\varphi_i = \varphi_j$, then it follows that $\varphi_i(\bar{1}) = 1i = i$ and $\varphi_j(\bar{1}) = 1j = j$ so $i = j$. Therefore this map is one to one. Let $\varphi_n \in \text{Hom}_{\mathbb{Z}}(\mathbb{Z}/n\mathbb{Z}, A)$ note that since A is a $\mathbb{Z}$ -module

□

Exercise 3 (10.2.5). *Exhibit all $\mathbb{Z}$ -module homomorphisms from $\mathbb{Z}/30\mathbb{Z}$ to $\mathbb{Z}/21\mathbb{Z}$.*

Proof.

□

SECTION 11.1

Exercise 4 (11.1.1). *Let $V = R^n$ and let $(a_1, a_2, \dots, a_n)$ be a fixed vector in V . Prove that the collection of elements $(z_1, z_2, \dots, z_n)$ of V with $a_1z_1 + a_2z_2 + a_3z_3 + \dots + a_nz_n = 0$ is a subspace of V . Determine the dimension of this subspace and find a basis.*

Proof.

□

Exercise 5 (11.1.2). *Let V be the collection of polynomials with coefficients in $\mathbb{Q}$ in the variable x of degree at most 5. Prove that V is a vector space over $\mathbb{Q}$ of dimension 6, with $1, x, x^2, \dots, x^5$ as a basis. Prove that $1, 1+x, 1+x+x^2, \dots, 1+x+x^2+x^3+x^4+x^5$ is also a basis for V .*

Proof.

